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ORIGINAL ARTICLE

Japanese subgingival microbiota in health vs disease and their roles in predicted functions associated with periodontitis

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Abstract

The present study aimed to identify and compare the microbial signatures between periodontally healthy and periodontitis subjects using 454 sequences of 16S rRNA genes. Subgingival plaque samples were collected from ten periodontally healthy subjects and ten matched chronic periodontitis patients. Bacterial DNA was extracted and next-generation sequencing of 16S rRNA genes was performed. The microbial composition differed between healthy subjects and periodontitis patients at all phylogenetic levels. Particularly, 16 species, including *Lautropia mirabilis* and *Neisseria subflava* predominated in healthy subjects, whereas nine species, including *Porphyromonas gingivalis* and *Filifactor alocis* predominated in periodontitis. UniFrac, a principal coordinate and network analysis, confirmed distinct community profiles in healthy subjects and periodontitis patients. Using predicted function profiling, pathways involved in phenylpropanoid, GPI-anchor biosynthesis, and metabolism of alanine, arginine, aspartate, butanoate, cyanoamino acid, fatty acid, glutamate, methane, proline, and vitamin B6 were significantly over-represented in periodontitis patients. These results highlight the oral microbiota alterations in microbial composition in periodontitis and suggest the genes and metabolic pathways associated with health and periodontitis. Our findings help to further elucidate microbial composition and interactions in health and periodontitis.

Keywords 16S rRNA · Bacteria · Oral microbiome · Periodontitis · Pyrosequencing

Introduction

Periodontitis is a common oral inflammatory disorder and a major cause of tooth loss in adults [1]. More than 700 bacterial species are reportedly found in the oral cavity [2], of which 400 species have been identified in periodontal pockets, and 100–200 species have been found in any particular individual [3, 4]. It is presently recognized that an imbalance of the normal microbiota, called dysbiosis, disrupts host homeostasis and causes periodontal tissue destruction [5, 6]. Although numerous uncultivable or unnamed species present in subgingival microbiota could contribute to the disease development or health maintenance, because of technical limitations, for decades the bacteriological etiology of periodontitis has

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been discussed focusing on several specific known species, such as in the red complex (*Porphyromonas gingivalis*, *Tannerella forsythia* and *Treponema denticola*) [7, 8]. Advances in sequencing technologies have enabled more comprehensive analyses of plaque microbiomes. Recent studies have been clarifying that the microbiota associated with periodontitis differs from that in periodontal health [9–13]. These studies showed that, together with recognized periodontal pathogens, several taxa including *Desulfobulbus* spp., *Filifactor alocis* and TM7 showed greater abundance in periodontitis and seemed to act as the core bacteria in the subgingival microbiome of disease [9, 14]. Metagenomic/metatranscriptomic analyses of subgingival dental plaque also showed the alteration of functions in diseased microbiota, and genes responsible for functions, such as bacterial motility, energy metabolism, and biosynthesis of lipopolysaccharides and nucleotide, were found to be over-represented in periodontitis [11, 15, 16]. The oral microbiota is also known to be modified by lifestyle habits such as smoking and diet [17], as well as systemic diseases such as diabetes, rheumatoid arthritis, and systemic lupus erythematosus [18, 19]. Furthermore, the composition of microbiota varies by ethnicity or region [20]. For example, clear differences in the salivary microbiome of orally healthy individuals were seen between South Korea and Japan, although the populations of both countries are composed primarily of the Asian mongoloid ethnicity [21].

Regarding the Japanese population, Yamashita et al. performed comprehensive microbiota analyses on saliva and the tongue [22, 23]. However, in the oral cavity, there are various areas with different local environments such as the gingival region, palatal plate, tongue dorsum, and tonsils [24]. Furthermore, the bacterial composition differs between saliva and subgingival pockets in periodontitis. We have previously examined the composition and characteristics of subgingival microbiota in periodontitis vs peri-implantitis, but we were unable to make comparisons to healthy counterparts [25, 26]. To the best of our knowledge, the analysis of microbiota in the periodontally healthy subgingival sulcus has not yet been conducted in the Japanese population. To identify bacterial etiologic factors of disease onset and progression, it is important to better define the characteristics of both the dysbiotic (imbalanced/diseased) and homeostatic (stable/periodontally healthy) microbiota.

The accumulation of plaque bacteria triggers inflammation that can disrupt host homeostasis, resulting in an increase in disease-associated species that contribute to periodontitis development [27]. The purpose of this study is to clarify the characteristics of subgingival microbiota in Japanese subjects exhibiting either periodontal health or periodontitis.

Materials and methods

Study subjects

Periodontally healthy and periodontitis subjects were recruited following a clinical oral examination at Tokyo Medical and Dental University Hospital. Inclusion criteria for all subjects were: (1) age ≥ 40 years; (2) > 20 teeth; (3) no use of systemic antibiotics, oral anti-microbial agents, immunosuppressive agents, steroids, anti-inflammatory drugs, and biological drugs within the past 3 months; (4) no professional tooth cleaning within the past month; (5) no previous periodontal treatment. Periodontally healthy subjects were defined as having a probing depth (PD) of ≤ 3 mm at all sites. Periodontitis was defined as at least four sites with PD ≥ 6 mm in each quadrant. PD and bleeding on probing (BOP) and PD ≥ 6 mm were recorded at all sampling sites.

Sample collection

Subgingival plaque samples were taken from the mesial sites of the upper first molar. All collected samples were from natural teeth, and there was no prosthesis or odontopathy. The sampling sites were isolated with sterile cotton rolls, and supragingival plaque was removed by cotton pellets. After drying the target sites, 10 paper points (#30; VDW, Munich, Germany) were inserted into the pocket/sulcus for 30 s. The samples were collected in a sterile tube and placed immediately in liquid nitrogen. They were stored at − 80 °C until further use.

Preparation of bacterial 16S rRNA gene amplicons

Bacterial DNA was extracted as described by Said et al. [28], with minor modifications. Briefly, the collected paper points were soaked in 1.5 mL PBS and 500 µL 10 mM Tris–HCl/10 mM EDTA buffer, and incubated with 500 µL 15 mg/mL lysozyme (Sigma-Aldrich, St Louis, MO, USA) for 1 h with gentle mixing. Purified achromopeptidase (Wako Pure Chemical Industries, Osaka, Japan) was added to a final concentration of 2000 units/mL, and samples were incubated for a further 30 min. Next, 10% of 100 µL sodium dodecyl sulfate (SDS) and 20 µL proteinase K (Merck, Darmstadt, Germany) were added to the suspension to final concentrations of 1% and 1 mg/mL, respectively, after which samples were further incubated for 1 h with gentle mixing. The paper points were then removed.

The lysate was treated with phenol/chloroform/isoamyl alcohol (Life Technologies Japan, Tokyo, Japan) and centrifuged at 15,000 × *g*. To precipitate the DNA, 100 µL 3 M

CH_3COONa and 1 mL isopropanol were added, and samples were centrifuged at 5000×*g* for 15 min. The DNA pellet was rinsed with 75% ethanol, dried, and dissolved in 500 µL TE. DNA was further treated with 1 mg/mL RNase A (Wako Pure Chemical Industries) for 30 min, precipitated by adding 1 mL 20% PEG solution (PEG6000-2.5 M NaCl), mixed gently on ice for 10 min, and then centrifuged at 15,000×*g* at 4 °C for 10 min. After discarding the supernatant, the DNA pellet was rinsed twice with 75% ethanol, dried, and dissolved in TE. The dissolved pellet was kept at 4 °C until just before use.

Sequence processing and analysis

The V1–V2 region of the 16S rRNA gene was amplified by polymerase chain reaction (PCR) with barcoded 27Fmod (5′-agrgtttgatymtggctcag-3′) and 338R primers (5′-tgctgcctcccgtaggagt-3′) as described previously [29]. Briefly, PCR was performed in 50 ml of 1×Ex Taq PCR buffer, forward and reverse primers (0.2 mM) and 20 ng template DNA. Thermal cycling consisted of initial denaturation at 96 °C for 2 min, followed by 25 cycles of denaturation at 96 °C for 30 s, annealing at 55 °C for 45 s and extension at 72 °C for 1 min, and final extension at 72 °C on a 9700 PCR system (Life Technologies Japan, Ltd.). PCR amplicons from the V1–V2 regions were purified by AMPure XP magnetic purification beads (Beckman Coulter, Brea, CA, USA), and quantified using the Quant-iT PicoGreen dsDNA Assay Kit (Life Technologies Japan). Equal amounts of each PCR amplicon were mixed and then sequenced using either 454 GS FLX Titanium or 454 GS JUNIOR sequencing platforms (Roche, Basel, Switzerland).

The sequences were analyzed by pipeline for pyrosequencing data from the 16S rRNA V1–V2 region. The data were screened and removed if either forward or reverse primers were missing. Then, after removing both primer sequences, the amplicons were further cleared by discarding all reads with average quality values below 25 and possible chimeric sequences. For taxonomy assignment and identification of chimeric sequences in the 16S rRNA gene data, we constructed our own databases from three publicly available databases: Ribosomal Database Project v. 10.27, CORE (http://microbiome.osu.edu/), and a reference genome sequence database obtained from the NCBI FTP site (ftp://ftp.ncbi.nih.gov/genbank/, December 2011) using the GLSEARCH software. Reads having BLAST match lengths < 90% with the representative sequence in three databases were considered as chimeras and removed. Finally, filter-passed reads were used for further analysis [28].

All 16S rRNA gene sequence data used in this study were deposited in DDBJ/GenBank/EMBL under accession number: DRA005979.

Operational taxonomic unit (OTU) clustering and UniFrac analysis

OTU clustering and UniFrac analysis from the filter-passed reads was performed based on 3000 randomly chosen high-quality reads/sample. The total reads were then sorted according to the average quality value and grouped into OTUs using UCLUST (http://www.drive5.com/) with a sequence identity threshold of 96%.

Characterisation of microbiota in co-occurrence networks

First, species with a relative abundance < 0.1% of total reads were removed [30, 31]. Co-occurrence coefficients were then calculated using the SparCC program [32] and 16S rRNA gene taxonomic abundances in each group. Ten iterations were used to estimate the median correlation of each pairwise comparison, and the statistical significance of each correlation was calculated by bootstrapping with 500 iterations [31]. Two-sided p and q values of < 0.05 and < 0.1 were considered statistically significant. Taxon pairs with SparCC values ≥ 0.4 were considered as exhibiting a co-occurrence relationship with a positive correlation. Our criterion for significance testing was more stringent than the previously used value of ≥ 0.3 [25]. Co-occurrence patterns were drawn using a network structure, in which each taxon and co-occurrence was indicated by a node and edge, respectively, for all species pairs with a positive correlation. The networks were visualized using Cytoscape software v.2.8 [33].

Predicting the functions of subgingival microbiota

Piphillin (http://secondgenome.com/solutions/resources/data-analysis-tools/piphillin/) was used to predict the functions of the 16S rRNA gene samples [34]. Piphillin generates metagenomic predictions from 16S rRNA data with 97% identity cutoff, based on the Kyoto Encyclopedia of Genes and Genomes (KEGG) reference database, release 82.0 (May 2017). Abundance values of metagenomic predictions were normalized by conversion into reads per kilobase of the transcript per million reads (RPM).

Statistical analysis

Clinical data and diversity indexes for periodontitis patients and healthy controls were compared by unpaired t test. The abundance of each bacterial taxon and RPM values obtained from the predicted metagenome data were compared by Mann–Whitney U tests. In this study, a value of $p < 0.05$ was considered statistically significant, except for cases of multiple comparisons. The p values were converted into q values by the Benjamini–Hochberg method as a measure of

the false discovery rate for multiple-comparison tests. We set the significance levels for the p and q values to < 0.05 and < 0.1, respectively. The value of '1−Spearman's coefficient' was used as the distance defining the difference between predicted functional genes. Analysis of similarities (ANOSIM) was used to test the significance of dissimilarity between periodontitis patients and periodontally healthy subjects. All statistical calculations were performed using R software (v3.1.1).

Results

Difference in subgingival microbiota between periodontally healthy and periodontitis subjects

A total of 10 periodontally healthy subjects and 10 age-matched chronic periodontitis patients were recruited to compare subgingival microbiota. Periodontitis patients had significantly more severe clinical parameters, including PD and BOP, compared to those of healthy controls (Table 1). A total of 219,036 sequences for the 16S rRNA gene (range 7796–14,693 per sample) were obtained after quality checking, and 13 phyla, 109 genera, and 385 species were identified. Results from rarefaction curves indicated that a sufficient number of reads was obtained (Supplementary Fig. S2). At phylum level, *Actinobacteria* and *Proteobacteria* were significantly more abundant in healthy controls, while *Spirochaetes, Synergistetes, Bacteroidetes*, and *Tenericutes* were significantly more abundant in periodontitis (Fig. 1a). Although the number of OTUs, Chao 1 estimates, and Shannon index did not show significant differences between healthy and periodontitis subjects (Supplementary Fig. S1), the microbial community of each group could be clearly separated on principal coordinate analysis (PCoA) plots. The ANOSIM test on UniFrac distances revealed the dissimilarity in the microbiome between healthy and periodontitis subjects ($R = 0.56$; $p = 0.002$) (Fig. 1b).

Table 1 Summary of clinical features among participants

Item	Healthy ($n = 10$)	Periodontitis ($n = 10$)	p
Age (year)[a]	60.3 ± 9.6	65.2 ± 13.1	0.4
Male/female	2/8	1/9	0.6
Number of teeth	28.1 ± 1.6	23.0 ± 2.7	< 0.01
PD (sampled site)[a, b]	2.3 ± 0.4 mm	6.5 ± 1.2 mm	< 0.01
BOP (sampled site)[c]	1+, 9−	10+, 0−	< 0.01

[a]Data were presented as mean ± standard deviation

[b]Pocket depth

[c]Bleeding on probing

Heat maps showing the relative abundance of genera and species that differed significantly between periodontally healthy and periodontitis subjects are presented in Fig. 2a and b. With analyses at the genus level, 11 genera, including *Rothia*, *Neisseria,* and *Leptotrichia*, showed higher abundance in health and 14 genera, including *Treponema, Eubacterium,* and *Tannerella,* showed higher abundance in periodontitis (Fig. 2a). With analyses at the species level, 16 species showed higher abundance in health and 17 species showed higher abundance in periodontitis (Fig. 2b). To further clarify the core microbiota in each group, we compared the core species in healthy and/or periodontitis subjects (Fig. 3).

Figure 3a shows species which were found in at least 80% of subjects and had a mean relative abundance > 1%. 16 species, including *Leptotrichia* sp., *Neisseria subflava*, *Lautropia mirabilis*, and *Rothia dentocariosa*, were more prevalent and abundant in healthy controls. 9 species, including *P. gingivalis*, *F. alocis*, *T. forsythia*, *P. endodontalis*, *P. denticola, and Prevotella tannerae*, were more prevalent and abundant in periodontitis patients. *F. nucleatum*, *Fusobacterium* sp., *Streptococcus* sp., *Parvimonas micra*, and *Prevotella oris*, were more prevalent and abundant in both groups.

Figure 3b shows species which were found in at least 80% of subjects yet only had a mean relative abundance < 1%. 12 species, including *Abiotrophia defectiva*, *Fusobacterium periodonticum*, and *Gemella morbillorum*, were prevalent in healthy controls. 11 species, including *Prevotella dentalis*, *Selenomonas sputigena* and *Treponema socranskii*, were prevalent in periodontitis patients. 10 species, including, *Actinomyces viscosus*, *Campylobacter gracilis*, and *Streptococcus mitis*, were prevalent in both groups. Among the species highly prevalent in both subjects, *Granulicatella adiacens* was significantly more abundant in healthy subjects and *Eubacterium brachy* was significantly more abundant in periodontitis subjects.

Co-occurring microbiome network in periodontally healthy and periodontitis subjects

To analyze co-occurrence relationships in 16S rRNA gene profiles of the microbiome, we constructed network structures, in which two co-occurring species were indicated by nods and connected by a degree. There were two and three main network(s) with more than four nodes in periodontally healthy and periodontitis samples, respectively. In these networks, means of 0.84 and 1.05 degrees per node connected 49 healthy and 40 periodontitis subjects' nodes, with clustering coefficients of 0.048 and 0.144, respectively (Fig. 4). Species accounting for > 80% of samples in each group were prevalent in all networks, with 31/49 and 17/40 such cases found in healthy and periodontitis samples, respectively. Networks in healthy samples showed that *Capnocytophaga*

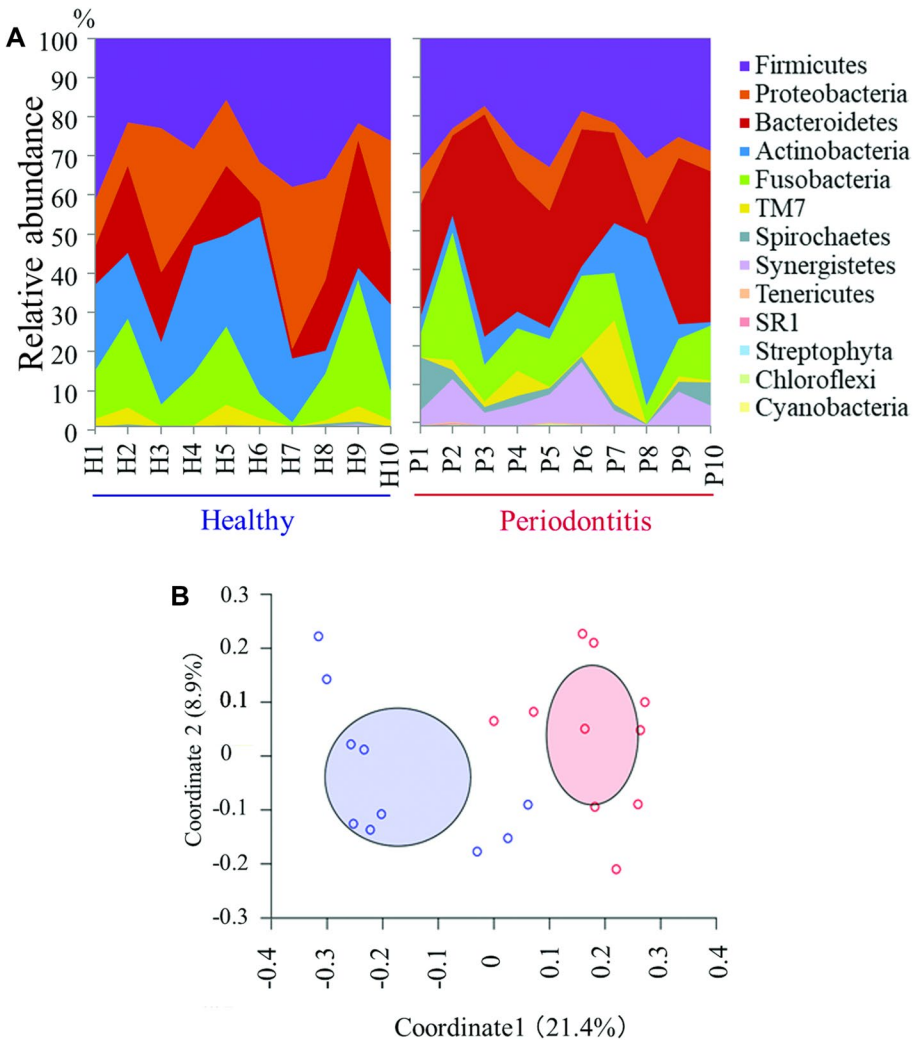


Fig. 1 Subgingival microbial composition in periodontally healthy and periodontitis subjects. **a** Microbial compositions at phylum level in healthy (blue dots) and periodontitis (red dots) subjects. **b** Comparison of subgingival microbiome between healthy and periodontitis subjects. Unweighted UniFrac analysis was used to generate distances between different samples and scattered plots were generated using PCoA. Healthy and periodontitis subjects are indicated by blue and red circles, respectively. The mean and standard deviation on each axis are indicated by an ellipse for each group

gingivalis, *Lautropia mirabilis*, *F. nucleatum* and *Streptococcus infantis* highly co-occured with other bacterial species (≥ 4°). On the other hand, the most complex and largest network module was found in periodontitis, which included well-established periodontopathic bacteria, such as *P. gingivalis*, *T. forsythia*, *F. nucleatum*, and *P. intermedia*.

Predicted function profiles of the microbiome in periodontally healthy and periodontitis subjects

To generate predicted metagenome profiles, 5172 and 4732 genes, and 254 and 252 pathways were assigned to healthy and periodontitis subjects, respectively. Accordingly, 830 and 788 genes were significantly more abundant in healthy and periodontitis controls, respectively, whereas 772 and 717 genes common to all samples were found in each of the two groups, respectively. Predicted KEGG pathways present in all healthy/periodontitis samples are shown in Fig. 5. Specifically, pathways related to two-component system, degradation of gamma-hexachlorocyclohexane, tetrachloethene and geraniol, photosynthesis, biosynthesis of pantothenate, CoA, ubiquinone and other terpenoid-quinone biosynthesis, nitrogen, cysteine and methionine metabolism, and peroxisome were significantly over-represented only in healthy subjects. In contrast, pathways involved in phenylpropanoid and GPI-anchor biosynthesis, as well as metabolism of alanine, arginine, aspartate, butanoate, cyanoamino acid, fatty acid, glutamate, methane, proline, and vitamin B6 were significantly over-represented only in periodontitis subjects. Based on dendrograms and PCoA plots, the predicted metagenome profiles of healthy and periodontitis subjects were confirmed by ANOSIM ($R = 0.49$; $p = 0.002$) (Fig. 6).

Discussion

Differences in bacterial community composition between periodontally healthy and periodontitis patients were clearly found at all taxonomic levels in the present study. Among those, principal species found in periodontitis included not

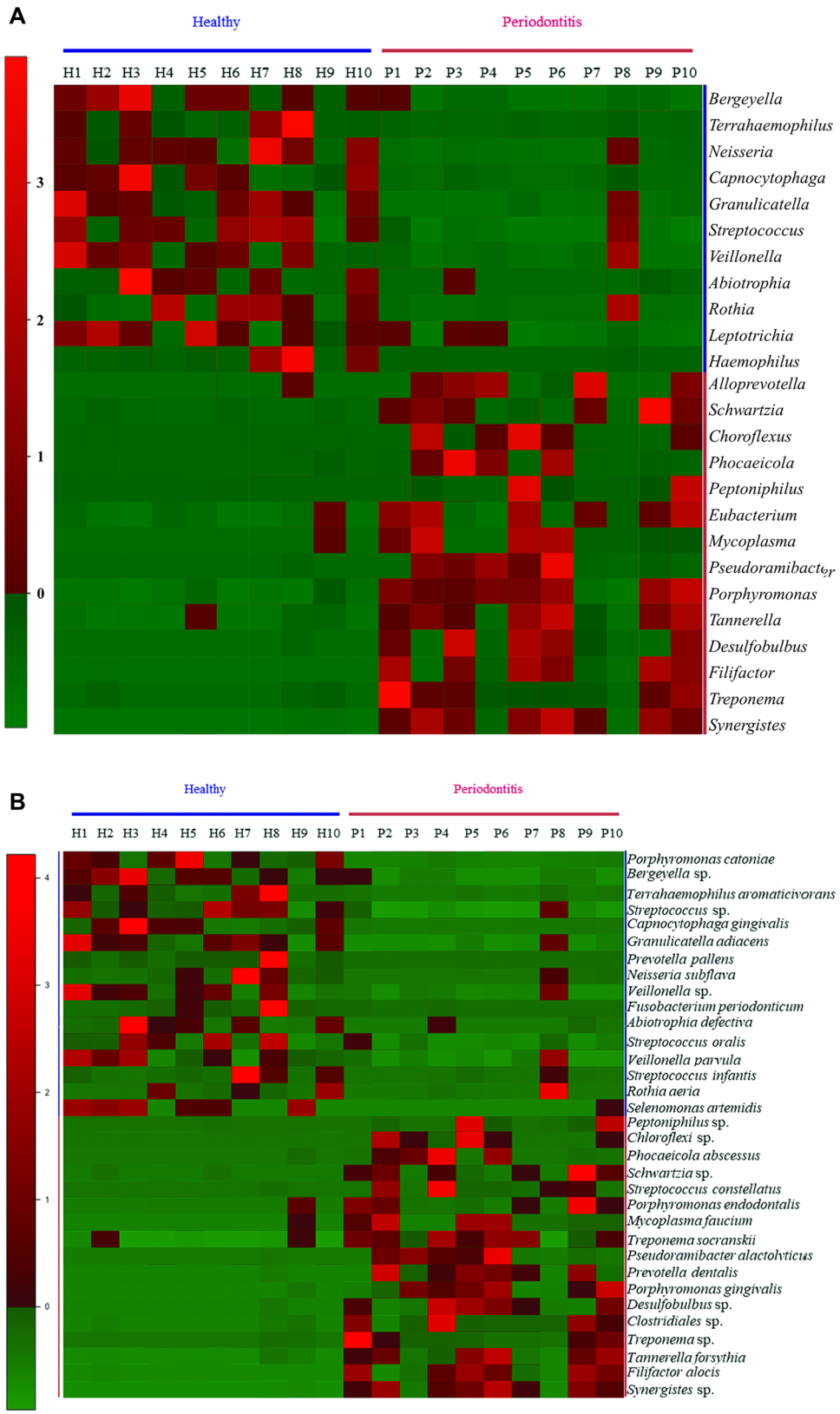


Fig. 2 Characteristics of bacteria from periodontally healthy and periodontitis subjects. **a** Relative abundances of genera that differed significantly between healthy and periodontitis subjects. Colors in the heat map reflect relative abundance from low (green) to high (red). **b** Relative abundances of species that differed significantly between periodontitis and healthy controls. Colors in the heat map reflect relative abundance from low (green) to high (red)

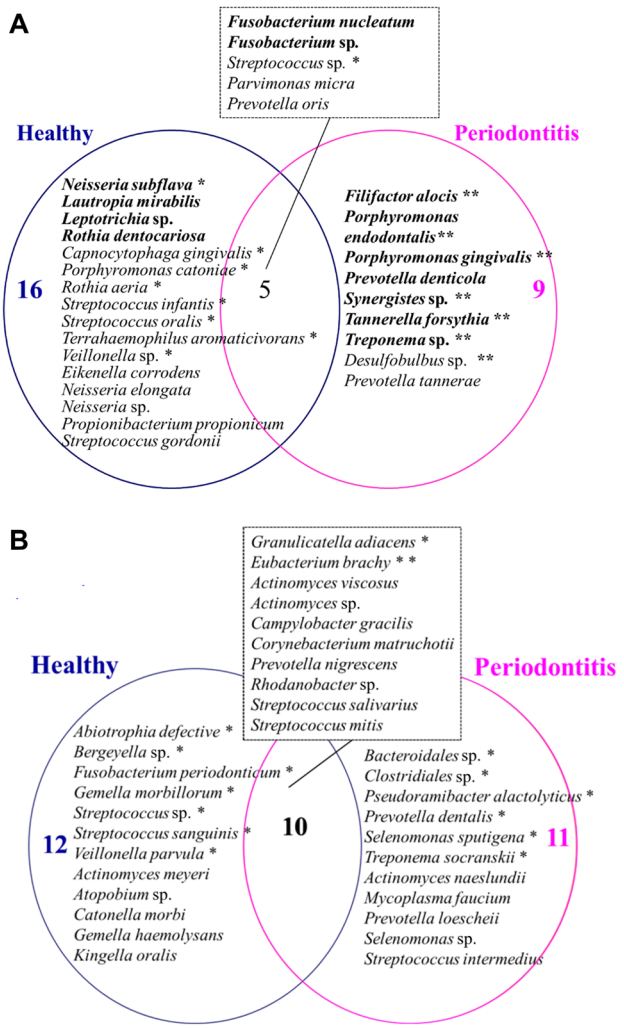


Fig. 3 Core microbiome of healthy and periodontitis subjects. **a** The model includes the species detected in healthy controls, periodontitis patients, and both, which were found in at least 80% of subjects and had a mean relative abundance > 1%. Species with a mean relative abundance > 2% are indicated in bold. * Significantly higher in periodontally healthy ($p < 0.05$ and $q < 0.1$); ** significantly higher in periodontitis ($p < 0.05$ and $q < 0.1$). **b** The model includes the species detected in healthy controls, periodontitis patients, and both, which were found in at least 80% of subjects and had a mean relative abundance < 1%. * Significantly higher in periodontally healthy ($p < 0.05$ and $q < 0.1$); ** significantly higher in periodontitis ($p < 0.05$ and $q < 0.1$)

only recognized periodontal pathogens, such as *P. gingivalis* and *T. forsythia*, but also newly identified pathogens [35]. One of them, *F. alocis*, has been proposed as a potential pathogen in periodontitis. Chen et al. [36] reported that *F. alocis*-centered co-occurrence group that consisted of eight phylotypes including *P. gingivalis* and *T. forsythia* was associated with periodontitis. In addition, the bacterium could promote pro-inflammatory cytokines as well as stimulate neutrophil degranulation and chemotactic activity in vitro [37].

Both Griffen et al. [9] and Szafranski et al. [38] have reported greater microbial diversity (Shannon index) and species richness (number of species/OTUs) at sites with periodontitis (PD ≥ 5 mm, PD ≥ 4 mm, respectively) compared to healthy controls. In contrast, in the present study, microbial diversity and species richness did not differ between periodontitis (PD ≥ 6 mm) and healthy control samples. Similar results were observed in a study by Kirst et al. [39], who compared microbial communities between periodontally diseased sites (PD ≥ 6 mm) and healthy control sites. They also compared these parameters at sites with 6-mm and 7–8-mm pockets. Interestingly, microbial diversity and species richness were significantly higher in 6-mm than in 7–8-mm pocket sites, and there was no significant difference in these parameters between 7 and 8-mm pocket sites and healthy sites. According to the 'Polymicrobial synergy and dysbiosis' theory, disease-associated bacteria increased in number in the healthy microbiota niche, resulting in a diverse community consisting of both disease-associated and beneficial bacteria [40]. Further progression of the disease alters this transitional state, leading to a more homogenous microbiota due to a clear increase of disease-associated bacteria. We considered that our results support the above-mentioned theory.

The analysis of specific microbial co-occurrence patterns is useful for identifying representative and important microbial associations in polymicrobial diseases [41]. In the present study, the most complex and largest network module was found in periodontitis, and it included many established periodontopathic bacteria, classified as red and orange complexes by Socransky et al. [42]. These established bacteria are known as late colonizers and are not observed in healthy-associated network structures. At the same time, the group also included recently identified bacteria as disease-associated species [14, 35]. Among them, *E. brachy* had the most interactions in periodontitis modules. Interestingly, despite its apparent importance and its increased presence in periodontitis patients, the relative abundance of *E. brachy* remained relatively low (< 1%). Even a comparatively lower abundance of bacteria can induce inflammatory diseases by transforming a symbiotic microbiota into a dysbiotic one [40, 43]. In keeping with that, we considered the possibility that even in low abundance, both established periodontopathic bacteria and newly identified disease-associated species such as *E. brachy* may together play key roles and contribute to an altered microbial community.

In this study, regarding the networks in healthy subjects, *C. gingivalis*, *L. mirabilis*, *S. infantis,* and *F. nucleatum* strongly co-occured with other species. Of these, *C. gingivalis, L. mirabilis,* and *S. infantis* were identified as core bacteria (highly prevalence and abundance) only in healthy microbiomes, and these finding support the results seen in previous reports [44–46]. On the other hand, *F. nucleatum*

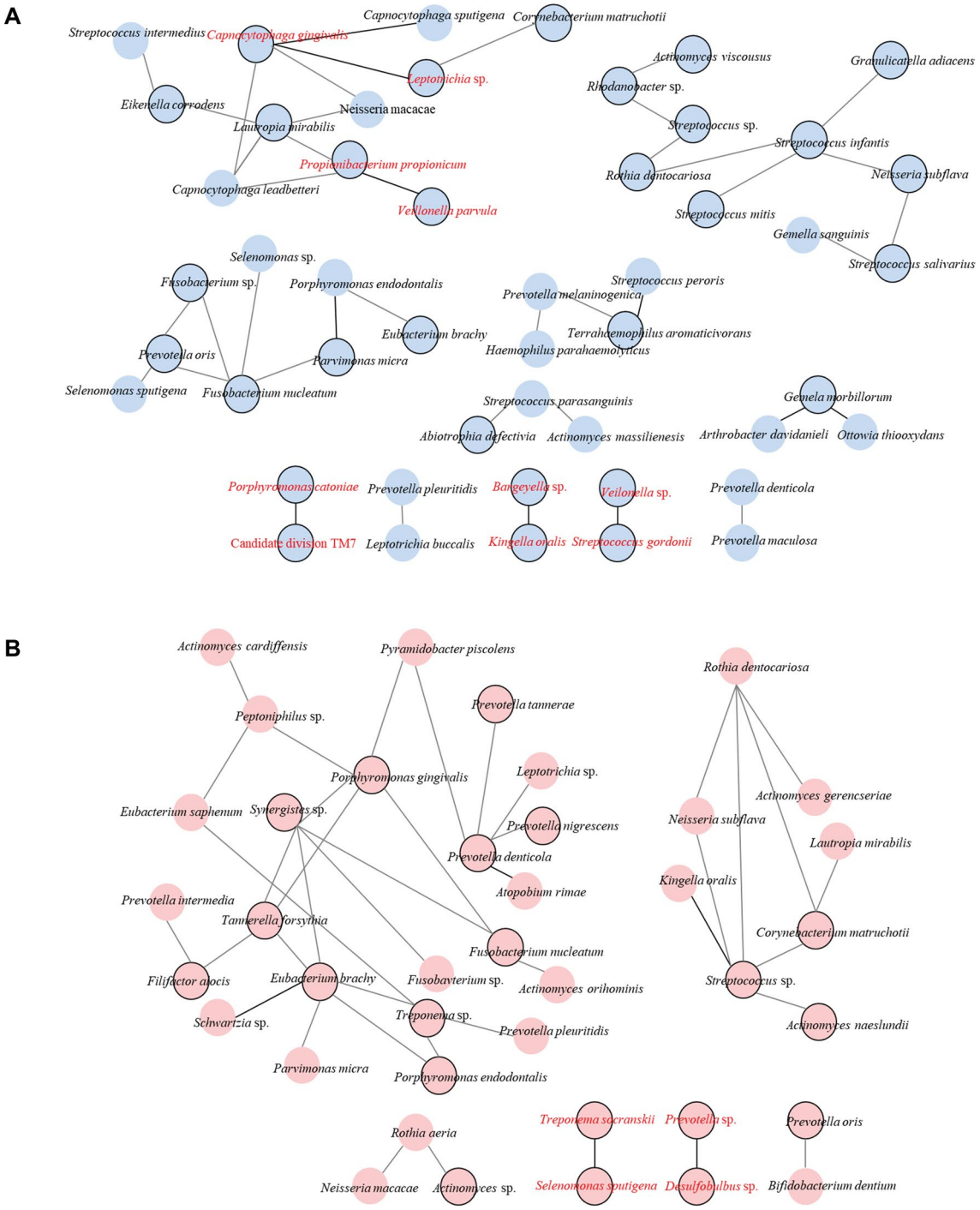


Fig. 4 Co-occurring network of microbiomes in periodontitis and periodontally healthy subjects. **a** Network module from healthy controls. **b** Network module from periodontitis subjects. Prevalent species (more than 80% of samples) are indicated with bold circles, and interactions with significant co-occurrence are indicated with bold lines ($p < 0.05$ and $q < 0.1$). Prevalent and significant species are indicated in red

is found by this study, and others [47], to be one of the core bacteria in both diseased and healthy microbiomes. Furthermore, this study showed that *F. nucleatum* is also highly associated with networks in periodontitis. Notwithstanding the above, in these networks, bacteria directly connected to *F. nucleatum* were found to be distinct in health vs periodontitis. We consider it possible that the altered interrelationships between *F. nucleatum* and other species may be

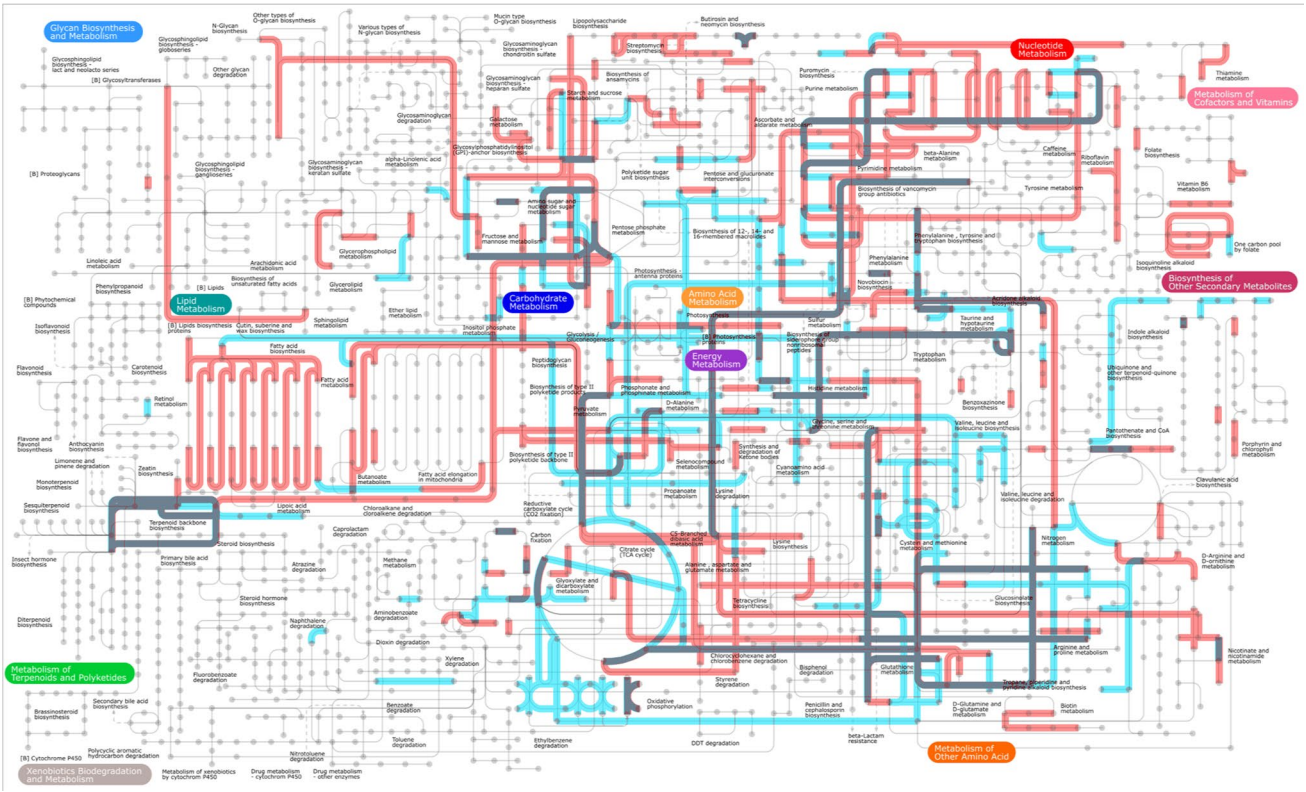


Fig. 5 Predicted metabolic pathways in healthy and periodontitis subjects. Blue lines indicate predicted KEGG pathways that were significantly over-represented in healthy subjects. Red lines indicate predicted KEGG pathways that were significantly over-represented in periodontitis patients. Purple lines indicate predicted KEGG pathways that were unchanged in health and disease

responsible for shifting the microbiota in a dysbiotic direction, from health to disease.

Examining predicted metagenome profiles can provide greater insight into functional differences between a healthy (symbiotic) and a periodontitis (dysbiotic) microbiome, although our data still need to be taken into consideration for the lack of actual function profile [11]. The algorithm Piphillin [48] was applied for this analysis and revealed that several potential pathways were significantly enriched in healthy and diseased sites. Liu et al. [49] have reported that a disease-associated microbiome was enriched in metabolic functions such as fatty acid metabolism and butanoate metabolism pathways. Enrichment of these two pathways was also confirmed in this study, although pathways for fatty acid biosynthesis were not enriched. Certain fatty acids are known to have anti-inflammatory effects, and an epidemiological study showed that higher dietary intakes of docosahexaenoic acid were associated with lower prevalence of periodontitis [50]. Furthermore, an in vitro study showed that oral mucosal lipids including fatty acids had an antibacterial effect against *P. gingivalis* [51]. Interestingly, oral mucosal and salivary fatty acids may be metabolized by the altered dysbiotic microbiota, resulting in a reduction of their protective function and consequent progression of periodontitis. Pathways related to the metabolism of arginine and glutamate were also more abundant in periodontitis patients. These amino acids play an important role in activating host immune responses [52, 53], and an increased metabolism of these compounds may induce disruption of host homeostasis. Moreover, Wang et al. [16] reported that the genes related to hydrogen sulfide metabolism were underrepresented in periodontitis. Hydrogen sulfide is a volatile sulfur compound (VSC), and clinical studies have shown that increased VSCs were positively correlated with the severity of periodontitis [54]. VSCs produced by oral anaerobic bacteria such as *P. gingivalis* and *P. intermedia* [55] not only cause halitosis but also induce apoptosis in periodontal cells [56]. Our results support the above by showing that pathways for metabolism of some VSC precursors (sulfur-containing amino acids; cysteine and methionine) were significantly over-represented in health. In addition, Jorth et al. [47] reconstructed metabolic networks using metatranscriptomic analysis. These networks represented five important pathways (nucleotide metabolism, butyrate metabolism, pyruvate, histidine degradation, and tetrahydrofolate metabolism) that were upregulated in periodontitis. Our predicted pathways data also showed that these five pathways were over-represented in periodontitis

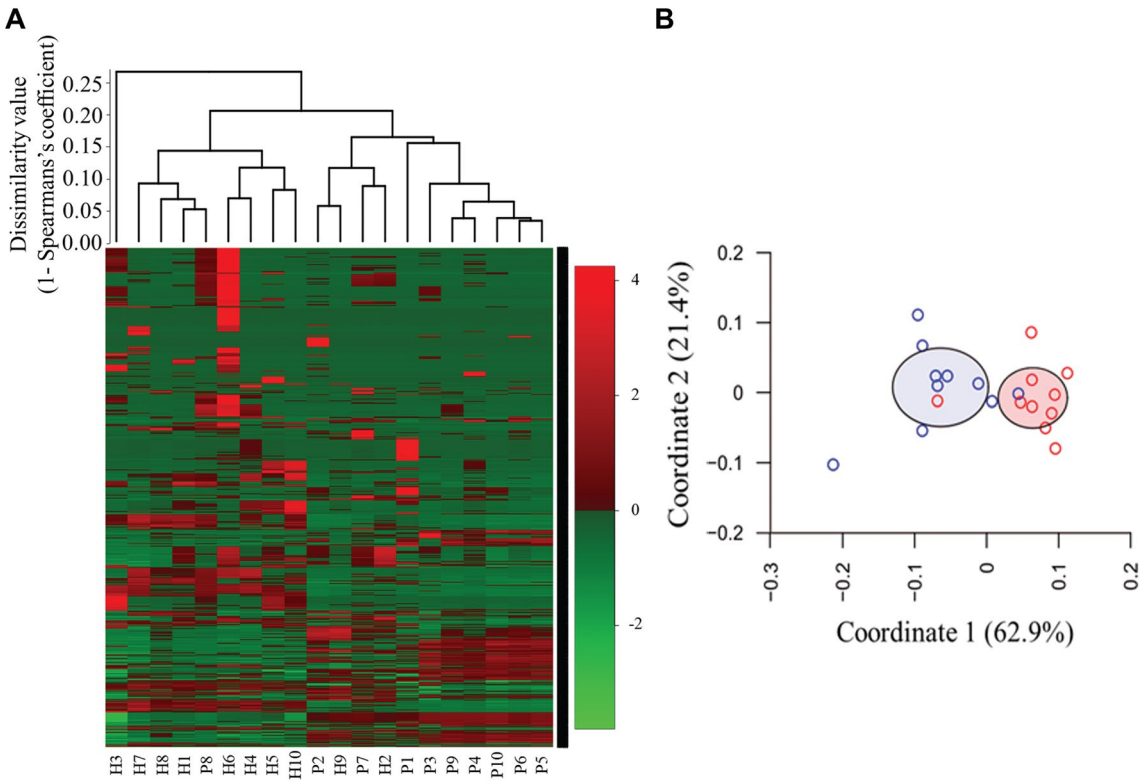


Fig. 6 Functional profiles from virtual metagenome analysis. **a** A dendrogram and a heat map of predicted abundance of each functional gene. Dissimilarity values for '1−Spearman's coefficient' were clustered using the average-linkage method, as shown in the dendrogram. **b** PCoA was performed for the dissimilarity matrix value of '1−Spearman's coefficient', and 10 samples each from periodontitis and periodontally healthy subjects are plotted with two coordinates. Periodontitis and periodontally healthy subjects are indicated by red and blue circles, respectively. The mean and standard deviation on each axis are indicated by an ellipse for each disease group

patients (Fig. 5). This further supports the usefulness of our predicted functions methods using 16SrRNA gene sequencing. While, we also have to note that the disadvantage of predicting functions based on species is that species can switch their metabolism based on, e.g., environmental conditions, and only measurements of RNA/protein/metabolites give a more realistic function profile.

This study could be considered limited by the use of a low number of patients and sampling of only one site for each patient. The availability of patients was limited by rigorous selection criteria. As for the use of single site sampling, we have demonstrated that all subgingival sites may show similar subject-specific microbiota in healthy patients [57], thereby reducing the need for multiple site sampling in health. As for single site sampling in the periodontitis patients, we extrapolated the adequacy of single site sampling for the study from the findings of the previously mentioned study.

In conclusion, our analysis of the 16S rRNA genes by pyrosequencing confirmed the difference in subgingival microbiota between periodontally healthy and periodontitis subjects, and enriched the understanding of the Japanese subgingival microbiome.

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Compliance with ethical standards

Conflicts of interest The authors declare that they have no conflict of interests.

Research involving human participants All procedures performed in studies involving human participants were in accordance with the ethical standards of Tokyo Medical and Dental University Institutional Review Board (No. 1138) and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed consent Written informed consent was obtained from all enrolled individuals.

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